

GREENHOUSE AUTOMATION

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EXISTING SYSTEM

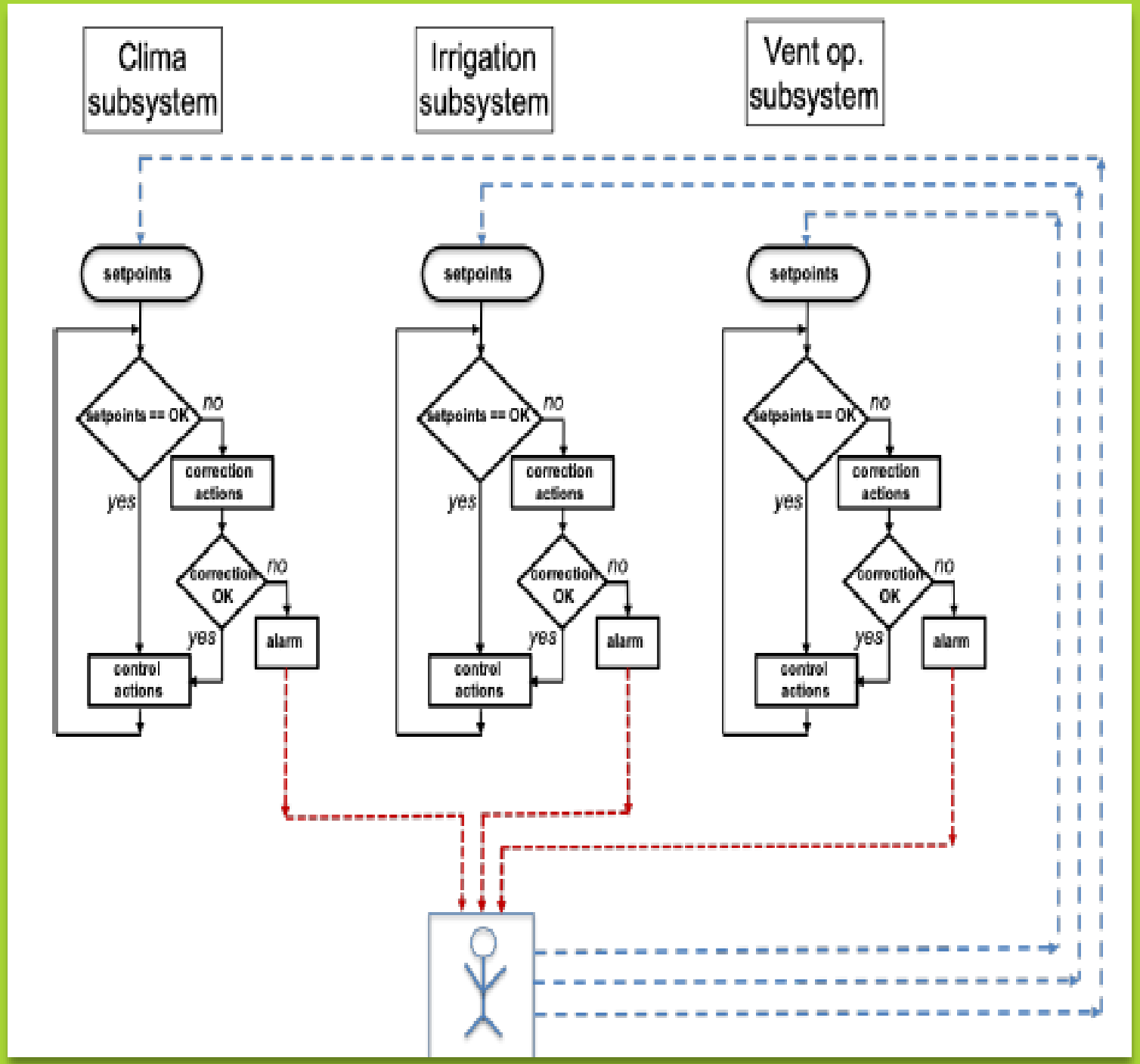
Automation is carried out with Monitoring and Controlling Techniques

Unit has an automated system that controls air conditioning, soil conditions and irrigation.

Control Sub sytems are not Integrated.

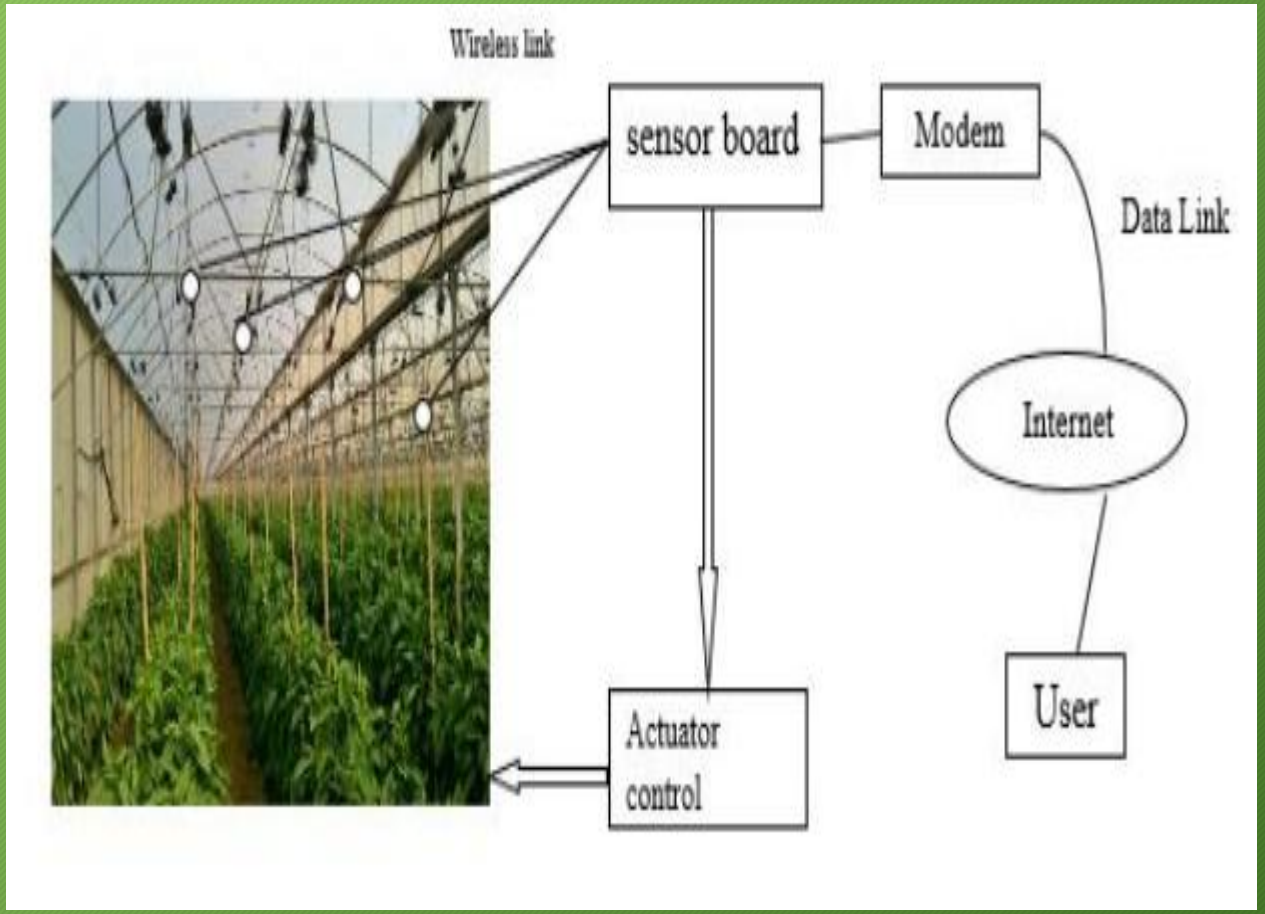
APPLIED TECHNOLOGIES

- Precision Agriculture - Automation in Agriculture
- Information and Communication Technologies(ICT)
- Sub – systems are connected to Artificial Intelligence Platform “Smarter Environments”



SENSOR NODES

- The main objective of the WSN system for greenhouse is to control the climatic conditions
- The sensor is designed for collecting information about climate of agriculture greenhouse like temp, pressure, light, humidity, CO2, wind speed and wind direction.



SMART GREENHOUSE REQUIREMENTS

- Crop – Sensors with data are considered as inputs
- Soil - Soil moisture, disease control should be treated
- Climate - Humidity and temperatures
- Water and Nutrients - Water Temperature, pH are variables
- Energy - Energy consumption and renewable generation
- User interfaces
- Events detection

SMART GREENHOUSE MODEL

HARDWARE

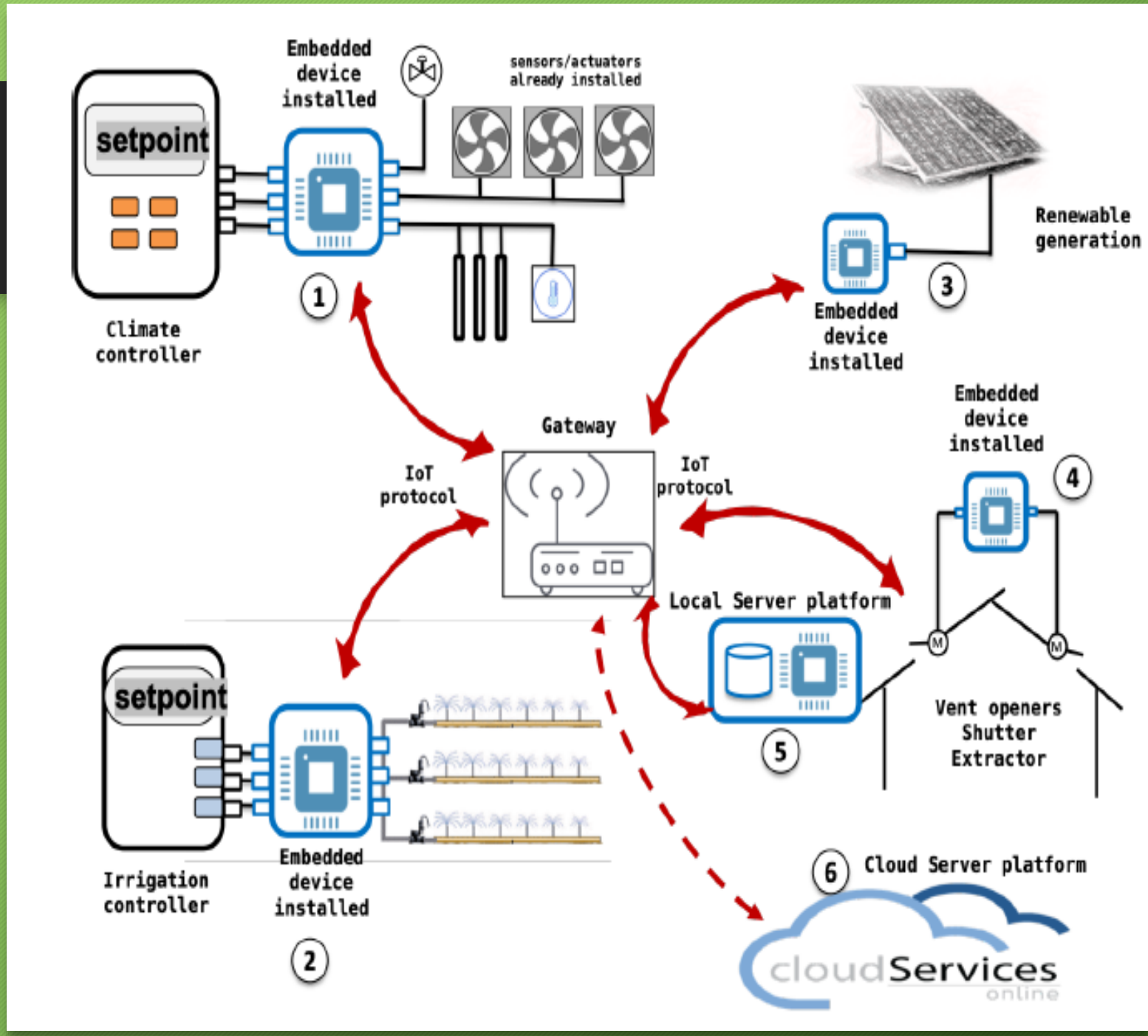
- Hardware Platform uses embedded devices to integrate greenhouses automated subsystems (climate, crop irrigation, vent openers and renewable energy generation)

SOFTWARE

- Software platform with artificial intelligence paradigms perform prediction and classification tasks to automate new actions.

SMART GREENHOUSE DESIGN

- Climate Controller
- Irrigation Controller
- Renewable Energy Generation
- Vent Openers Shutter
- Local Server Platform
- Cloud Server Platform



LEVELS OF SMART GREENHOUSE DESIGN

First level - Formed by control nodes 1,2,3,4

- The controllers connect to each other and to the local network, managed by a gateway

Second level - Formed by control Nodes 5 and 6

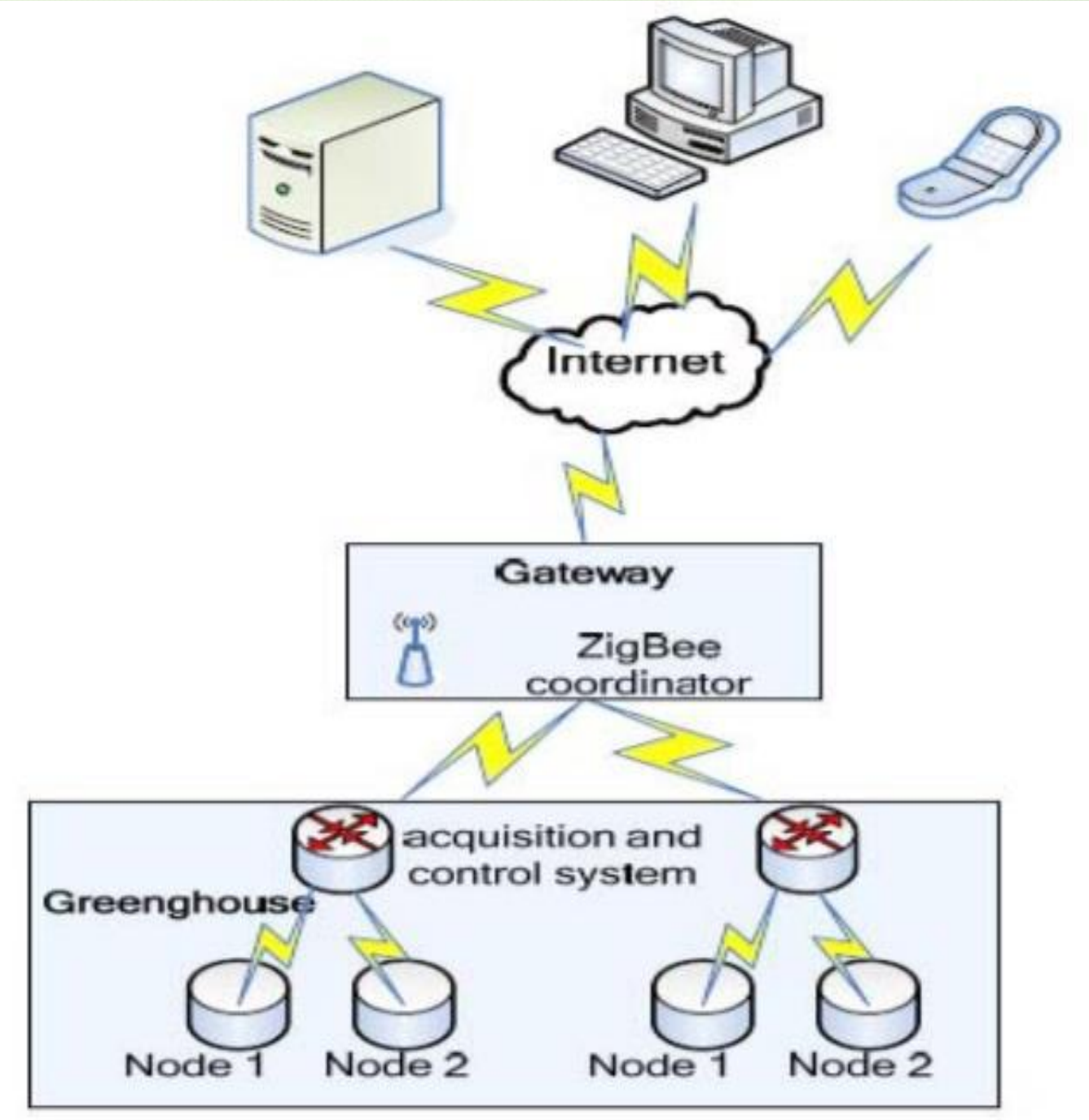
- The controlling nodes communicate with the network management servers using IoT protocols

✓ IoT Communication

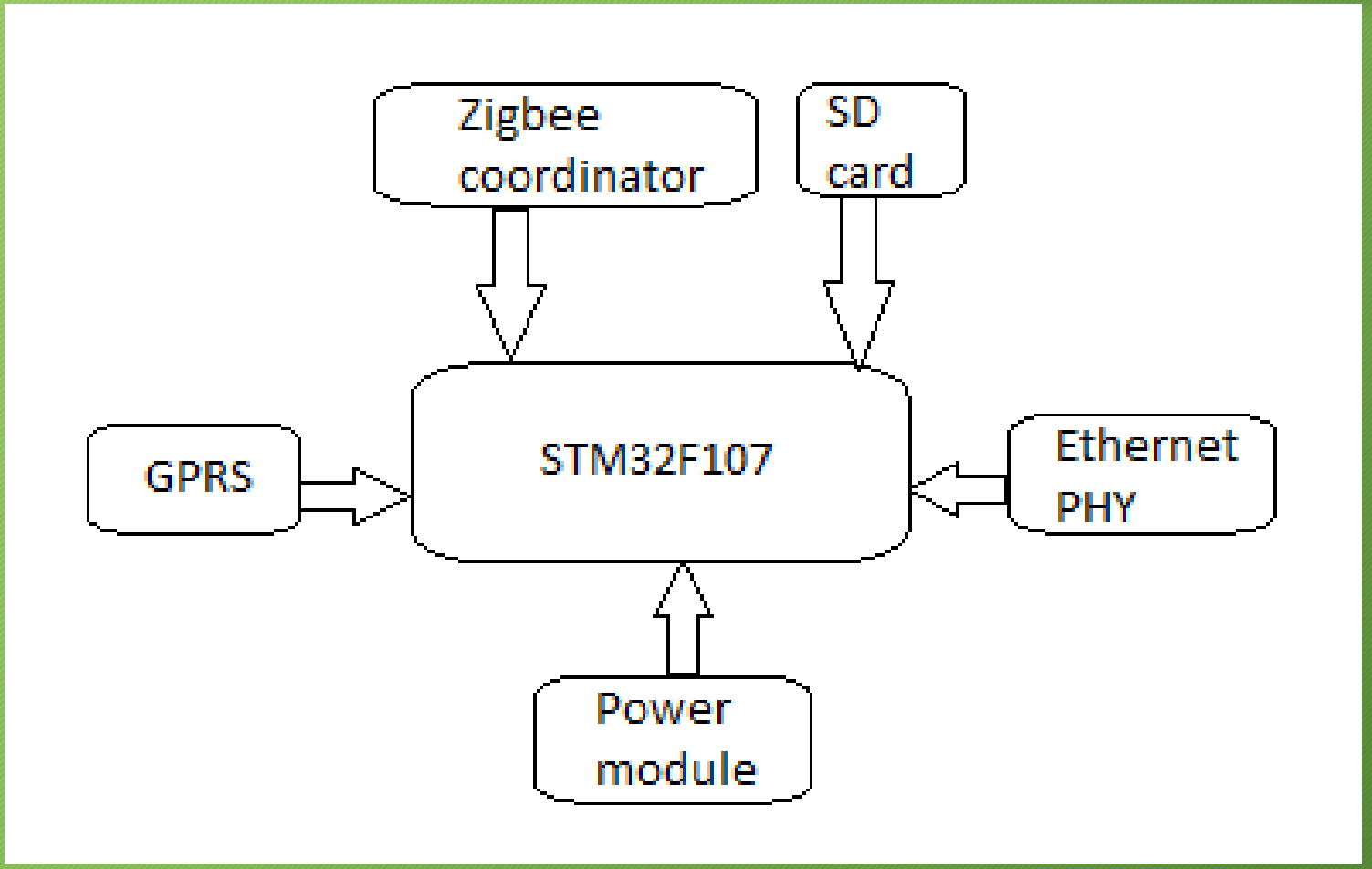
✓ Processing Capacity

IOT COMMUNICATION

- CC2530 - Processing chip of the wireless sensor nodes
- The ZigBee technology is adopted in wireless communication, the gateway uses the processor and Linux operating system as the core.



DESIGN OF IOT GATEWAY

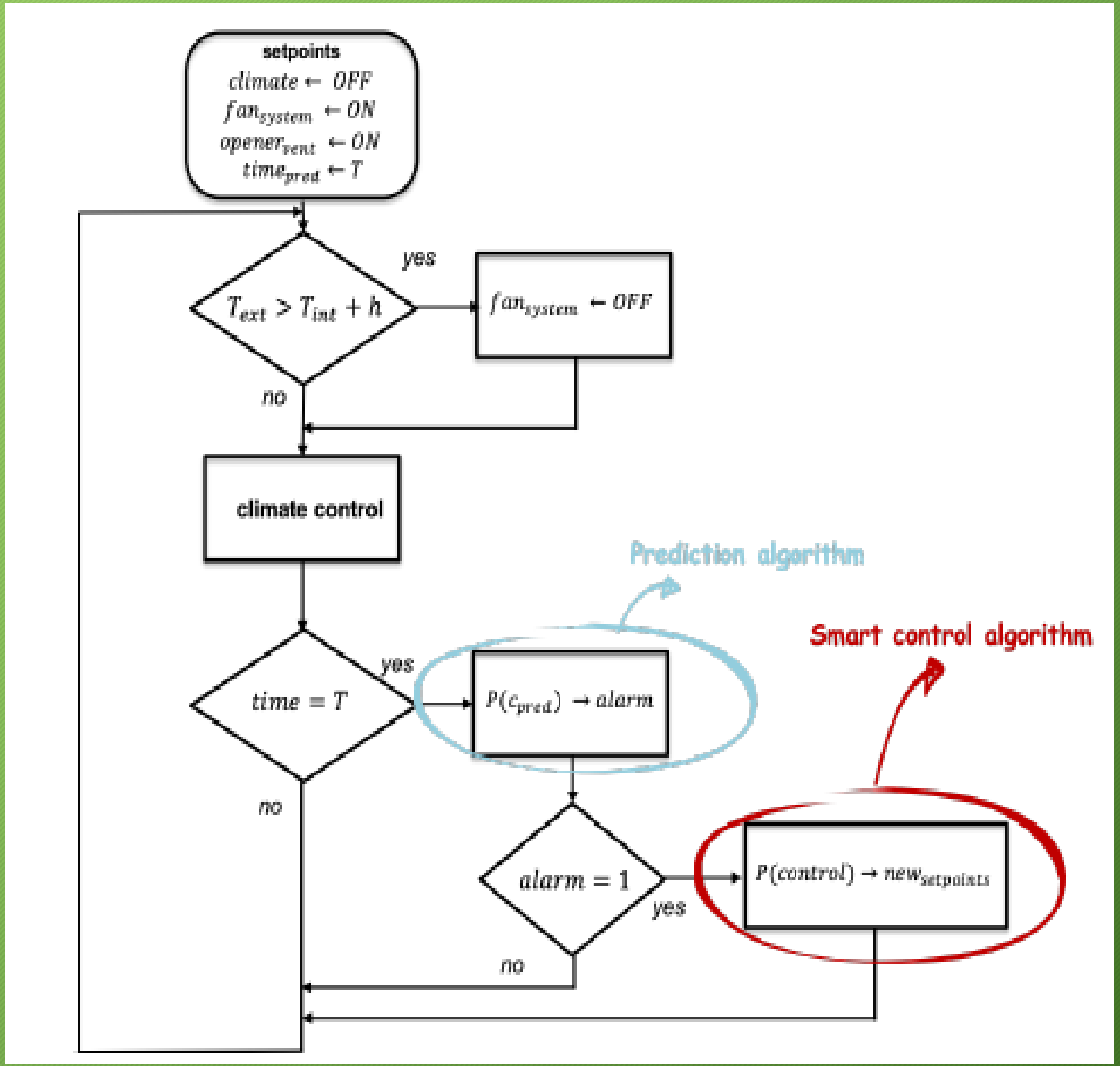


AI IN GREENHOUSE AUTOMATION

- Algorithms that implement AI models are installed in embedded devices with multi-threaded operating system
- The programming language used is Python
- The set points are defined in hourly sections depending on the crop and the expected climatic conditions.

PREDICTION INTEGRATION AND SMART CONTROL

- A new prediction algorithm has been developed
- The User Centered Design(UCD) method is followed where both the agronomist and the ICT technician participate in this development, services are designed, implemented and validated



RANDOM FOREST REGRESSION

- The agronomist schedules the conditions of each of them manually, using a web interface.
- The algorithm predicts the expected temperature for every 15 minutes, With this prediction the system can act and make control decisions.
- The results of temperature prediction obtained with the model proposed are given by RF (Random Forest) Algorithm.

Results of RF Algorithm

	Texterior	HRexterior	Vviento	Dviento	Radiacion	HRinterior	Tinterior
0	13.581	93.238	3.06870	43.144	1.3312	59.931	24.044
1	13.500	93.638	1.35620	43.137	1.2812	59.819	23.969
2	13.531	93.856	0.64375	54.394	1.3125	59.756	23.900
3	13.581	93.906	0.43750	58.294	1.3313	59.638	23.950
4	13.544	93.963	0.20625	69.912	1.3250	59.550	24.000

Input:
dataset.csv

1



Random Forests Library

```
In [92]: from sklearn.ensemble import RandomForestRegressor

In [93]: forest = RandomForestRegressor(n_jobs=2, oob_score=True, n_estimators=10000)
         forest.fit(X,Y)

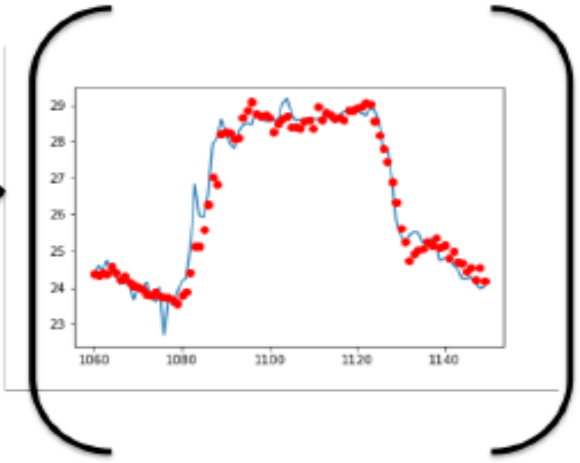
Out[93]: RandomForestRegressor(bootstrap=True, criterion='mse', max_depth=None,
                    max_features='auto', max_leaf_nodes=None,
                    min_impurity_decrease=0.0, min_impurity_split=None,
                    min_samples_leaf=1, min_samples_split=2,
                    min_weight_fraction_leaf=0.0, n_estimators=10000, n_jobs=2,
                    oob_score=True, random_state=None, verbose=0, warm_start=False)
```

Controller Process:
random forest algoritm

2

3

Ouput:
Tint.
prediction



CONCLUSION

- The current state of the technologies in greenhouse automation is revised
- Expert agronomist knows the operation rules, ICT technologies, such as embedded hardware, communication protocols (IoT), algorithm processing with high-level languages (Python) and artificial intelligence models (random forest regression)

“Convert greenhouses into Smart environments”

REFERENCES

- **Francisco Javier Ferrández-Pastor , Sara Alcañiz-Lucas, Juan Manuel García-Chamizo and Manuel Platero-Horcajadas -** *Smart Environments Design on Industrial Automated Greenhouses*
- Ferrández-Pastor, F.J.; García-Chamizo, J.M.; Nieto-Hidalgo, M.; Mora-Martínez, J. Precision Agriculture Design Method Using a Distributed Computing Architecture on Internet of Things Contex. Sensors **2018**, 19,603. doi:10.3390/s18061731
- **Harshada D Thite, S V. Athawale** T.E. - An IoT based Greenhouse System using ZigBee Technology

THANK YOU